

FINAL REPORT

Credit Card Fraud Detection Using AI & Machine Learning

Domain: Financial Services / Machine Learning

Github Link: <https://github.com/peepoalparce/AIM-AI-ML-Capstone-Project.git>

1) Problem Understanding & Framing

Business Problem

Financial institutions face significant financial losses and operational overhead due to fraudulent credit card transactions. Fraudulent transactions are extremely rare (0.17% of the dataset's total volume), making early detection challenging. The primary business objective is to identify fraudulent transactions in real time, reducing direct financial exposure while minimizing customer friction caused by false transaction declines.

Data Science Problem

The goal of this project is to build a supervised binary classification model that accurately distinguishes fraudulent credit card transactions (Class = 1) from legitimate ones (Class = 0). Since fraud cases represent an extreme minority class, the model must learn meaningful patterns without overfitting to noise, maximizing recall while maintaining operational precision.

Success Metrics & KPIs

- *PR-AUC (Primary Metric)*: The Precision-Recall Area Under the Curve is the primary technical benchmark since the accuracy is misleading on severely imbalanced datasets.
- *Recall (Secondary Metric)*: Measures the proportion of actual fraud cases caught, directly reducing financial losses.
- *Precision (Operational Control)*: Ensures that flagged transactions are genuinely fraudulent, limiting unnecessary customer service interventions and card blocks.
- *Business KPIs*: Estimated cost savings from intercepted fraud, reduction in undetected fraud cases, and an operationally manageable alert rate.

2) Data Collection & Understanding

Dataset Used: Credit Card Fraud Detection (Kaggle)

Link: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Dataset Overview

- The dataset contains 284,807 credit card transactions with 31 features, including anonymized PCA-transformed variables and a binary fraud label.
- Class Distribution: 284,315 legitimate transactions (99.827%) vs. 492 fraud cases (0.173%).
- Data Quality: 0 missing values; 1,081 duplicate rows identified and removed during data cleaning.

Structured Data Dictionary

Variable Name	Data Type	Range / Distribution	Description & Measurement Units
Time	Continuous (Float)	0 to 172,792.00	Seconds elapsed since the first transaction in the dataset.
Amount	Continuous (Float)	0 to 25,691.16	Transaction monetary value in USD/EUR.
V1 – V28	Continuous (Float)	Unbounded (PCA Scaled)	Anonymized features derived from Principal Component Analysis (PCA).
Class	Categorical (Binary)	0 or 1	Target variable: 0 = Legitimate, 1 = Fraudulent.

3) Data Preprocessing, Applied EDA & Feature Engineering

Preprocessing & Clean Data

- **Duplicate & Outlier Management:** Removed 1,081 duplicate records. Detected 31,904 outliers in Amount via the IQR method; these were retained as high-value transactions often carry crucial fraud risk signals.
- **Feature Scaling:** Applied StandardScaler to Time and Amount to normalize feature magnitudes alongside the pre-scaled PCA components.

Feature Engineering & Domain Experiments

To demonstrate domain-driven feature exploration on anonymized data:

- **Log Transformation:** Created $\text{Log_Amount} = \log(\text{Amount} + 1)$ to stabilize the severe right-skewness of transaction amounts.
- **Risk Tiering:** Engineered a binary indicator $\text{Is_High_Amount} = 1$ if $\text{Amount} \geq \$100$, serving as a proxy flag for high-value fraud auditing.
- **Feature Selection:** Embedded selection via Random Forest (SelectFromModel) identified top-contributing features including V2, V4, V10, V12, V14, V16, and V17.

Applied EDA & Dimensionality Reduction

- **Distribution Analysis:** Confirmed extreme class imbalance and evaluated feature correlations.
- **Clustering Analysis:** Visualized feature space using PCA and t-SNE projections, demonstrating partial non-linear separation between fraudulent and non-fraudulent clusters.

4) Model Implementation & Operational Comparison

Implemented several supervised machine learning models to classify transactions as either fraudulent or legitimate. Given that the dataset contains explicit labels (Class = 0/1), supervised learning is the appropriate approach. Three models commonly used in fraud detection and well-suited for tabular, imbalanced data were selected:

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier

Comparative Performance Matrix

Model	Fraud Precision	Fraud Recall	Fraud F1-Score	False Positives (FP)	False Negatives (FN)	ROC-AUC	PR-AUC
Logistic Regression	0.052326	0.918367	0.09901	1,630	8	0.971987	0.73347
Random Forest	0.81	0.826531	0.818182	19	17	0.969531	0.868734
XGBoost	0.237057	0.887755	0.374194	280	11	0.977454	0.853586

Random Forest was selected as the optimal production model. While Logistic Regression achieved the highest recall (91.84%), it generated 1,630 false positives, which would cause unacceptable operational overhead and block legitimate customer purchases. Random Forest achieves a superior PR-AUC (0.8687) and limits false positives to just 19 cases out of 56,864 transactions, striking the ideal trade-off between fraud capture and customer experience.

5) Critical Thinking: Ethical AI, Bias Auditing & Risk Mitigation

Data Leakage Prevention

To strictly prevent data leakage, SMOTE oversampling was applied exclusively to the training set (X_train) after performing the stratified train-test split. The holdout test set (X_test) remained completely untouched and un-sampled, ensuring test metrics accurately reflect performance on real-world imbalanced data distributions.

Overfitting & Model Validation

Hyperparameter optimization was executed using 5-Fold Stratified Cross-Validation on the training set, tuning structural constraints (n_estimators, max_depth, min_samples_split) to prevent decision trees from memorizing minority class instances.

Ethical AI & Algorithmic Bias Audit

Since demographic attributes were unavailable, transaction amount was used as a proxy grouping variable (High Amount $\geq$ \$100 vs. Low Amount $<$ \$100) to evaluate algorithmic equity across financial tiers:

- High-Amount Recall: 0.8222
- Low-Amount Recall: 0.8302
- Demographic Parity Ratio: 0.8902
- Equalized Odds Difference (TPR): 0.0080
- Equalized Odds Difference (FPR): 0.0000
- Disparate Impact: 0.8902

The audit confirms minimal performance disparity between transaction values, verifying that the model adheres to principles of operational fairness, transparency, and non-discrimination.

Explainability Suite

To ensure compliance with financial regulations, feature influences were validated using Permutation Importance, Partial Dependence Plots (PDP/ICE) for V14, and LIME local explanations for individual flagged transactions.